

EDUCATION

Georgetown University

Ph.D. in Computer Science

– Advisor: Prof. Alexander Golovnev

Washington DC, USA

Fall 2023 and Undergoing

Georgetown University

Masters in Computer Science, CGPA: 3.79/4.00

– Advisor: Prof. Alexander Golovnev

– Thesis: “Circuit Lower Bounds via Substitutions”

– Committee: Alexander Golovnev, Bala Kalyanasundaram, Muthuramakrishnan Venkitasubramaniam

Washington DC, USA

Fall 2021–Spring 2023

College of Engineering, Pune

B.Tech in Metallurgy and Materials Science Engineering, CGPA: 6.68/10

– Advisor: Prof. K. R. Kamble

– Thesis: “Piezoelectricity in AlN bulk ceramics”

Pune, India

2017–2021

RESEARCH INTERESTS

Complexity Theory, Cryptography, Circuit Lower Bounds, Randomness Extraction.

PUBLICATIONS

Efficient strong 2-source non-malleable extractor for any linear min-entropy

Divesh Aggarwal, Pranjal Dutta, Saswata Mukherjee, **Satyajeet Nagargoje**, Maciej Obremski

The 45th Annual International Cryptology Conference (**CRYPTO 2025**)

Santa Barbara, USA

Hilbert Functions and Low-Degree Randomness Extractors

London, UK

Alexander Golovnev, Zeyu Guo, Pooya Hatami, **Satyajeet Nagargoje**, Chao Yan

The 28th International Conference on Randomization and Computation (**RANDOM 2024**)

Range Avoidance for Constant-Depth Circuits: Hardness and Algorithms

Atlanta, USA

Karthik Gajulapalli, Alexander Golovnev, **Satyajeet Nagargoje**, Sidhant Saraogi

The 27th International Conference on Randomization and Computation (**RANDOM 2023**)

RESEARCH VISITS

National University of Singapore (NUS)

Singapore

Research Assistant

June 2024- November 2024

– Worked on showing that a random univariate degree $O(n)$ polynomial over F_{2^n} is an exponentially small error 2-Source Extractor (as well as Non-Malleable 2-Source Extractor) for all linear entropy sources. This work is currently under submission.

– Host: Prof. Divesh Aggarwal.

Ohio State University

Research Assistant

Columbus, OH, USA

May 2023- August 2023

- Worked on showing that a random polynomial is an Extractor for distributions sampled by low-degree circuits using Hilbert Functions. This result appeared at RANDOM-2024 titled “Hilbert Functions and Low-Degree Randomness Extractors”.
- Host: Prof. Pooya Hatami.

Georgetown University

Research Assistantship

Washington, DC, USA

Dec 2021- Jun 2023

- Proved circuit lower bounds in $AC^0[p]$ model for the boolean function MAJORITY using a novel measure called substitution complexity of a function. This work appears in the thesis titled: “Circuit Lower Bounds via Substitutions”.
- Worked on showing the hardness of local Range Avoidance Problem as well as finding algorithms in the various regimes of the problem. This work appeared at RANDOM-2023 titled “Range Avoidance for Constant-Depth Circuits: Hardness and Algorithms”.
- Advisor: Prof. Alexander Golovnev.

TEACHING ASSISTANTSHIP

COSC-3200: Introduction to Algorithms

Fall 2025

Georgetown University

COSC-3200: Introduction to Algorithms

Spring 2025

Georgetown University

COSC-030: Math Methods for Computer Science

Fall 2023

Georgetown University

COSC-240: Introduction to Algorithms

Fall 2022

Georgetown University

COSC-540: Graduate Algorithms

Fall 2022

Georgetown University

SELECTED TALKS

Efficient randomized strong 2-source non-malleable extractor for any linear min-entropy

February 2026
Baltimore, USA

Johns Hopkins University

Efficient randomized strong 2-source non-malleable extractor for any linear min-entropy

February 2026
Ithaca, USA

Cornell University

Efficient randomized strong 2-source non-malleable extractor for any linear min-entropy

November 2025
Washington DC, USA

Georgetown University

Efficient randomized strong 2-source non-malleable extractor for any linear min-entropy

October 2025
Austin , USA

Short talk at DavidFest, held at the University of Texas at Austin

Hilbert Functions and Low-Degree Randomness Extractors Tata Institute of Fundamental Research (TIFR)	December 2024 Mumbai, India
Hilbert Functions and Low-Degree Randomness Extractors National University of Singapore (NUS) Theory Seminar	October 2024 Singapore
Hilbert Functions and Low-Degree Randomness Extractors RANDOM 2024, London School of Economics	September 2024 London, UK
Range Avoidance for Constant Depth Circuits: Hardness and Algorithms Georgetown University	April 2023 Washington DC, USA

REFERENCES

Dr. Alexander Golovnev Assistant Professor Department of Computer Science at Georgetown University alex.golovnev@gmail.com	Washington, DC, USA
Dr. Pooya Hatami Assistant Professor Department of Computer Science at The Ohio State University (OSU) pooyahat@gmail.com	Columbus, OH, USA